The Readout Condition

Distinguishability, Access, and Epistemic Warrant

Yaoharee Lahtee[1]

[1]Independent Scholar, Bangkok, Thailand, ORCID: 0009-0005-3861-0626

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Abstract

Claims routinely distinguish more finely than the access sources invoked on their behalf. The central epistemic problem is not merely whether the resulting claim is true or justified, but whether each distinction in it is correctly attributed to the source, model, background evidence, relevance restriction, or decision rule that actually supplied the discrimination. This paper develops the *Readout Condition* as a necessary architecture for source-relative epistemic warrant. Its core is a three-part norm: *existence* (every claim-level distinction has an epistemic provenance path), *attribution* (a distinction credited to a source must be licensed by that source), and *disclosure* (essential augmentations must remain visible to audit). In the exact functional case, source attribution requires constancy on readout fibers and hence factorization through the readout. For non-transitive indiscriminability, the correct extension is pointwise: a claim token at an actual alternative is licensed only if its local indiscriminability neighbourhood lies inside the claim-value fiber, avoiding an illicit transitive closure. In stochastic settings, Blackwell garbling provides an *access audit* but cannot by itself detect assumption-driven silent lifts; a complementary identification ladder tracks the weakest augmentation layer at which a distinction becomes identified. The resulting typed provenance DAG supports a defeater-routing result: source-specific defeaters propagate exactly along the distinctions that depend on that source, while misattribution misroutes defeat and correction. Classical and pre-modern pressure tests from Dignāga–Dharmakīrti, Madhyamaka, Avicenna, and Suhrawardī constrain the framework's neutrality. The proposed contribution is not a new factorization theorem, data-processing inequality, or general notion of provenance, but a unified distinction-level architecture linking access, identification, attribution, disclosure, and epistemic revision.

1. Introduction

Epistemic practices routinely transform limited access into specific claims. A thermometer produces a reading; a clinician receives a test result; a witness reports an event; a classifier emits a label; an institution converts records into a decision. In each case, a later claim may distinguish more finely than the immediate readout. Such sharpening is often legitimate. The difficult question is not whether all downstream specificity is forbidden, but *what supplied it and whether it is attributed correctly.*

This paper calls the governing requirement the **Readout Condition**. The mature form defended here has three logically distinct components.

Principle 1 (Existence)**.** *Every claim-level distinction that bears epistemic weight must have an epistemic provenance path.*

Principle 2 (Attribution)**.** *If a claim attributes a distinction to a source S, then S must license that distinction under the relevant access condition.*

Principle 3 (Disclosure)**.** *If a distinction depends essentially on an augmentation beyond the attributed source, that augmentation must remain declared or recoverable in an epistemic audit.*

The conjunction is the Readout Condition. It is stronger than a generic demand for provenance and weaker than a complete theory of knowledge. It permits theory, priors, models, testimony, background evidence, relevance restrictions, and decision policies; what it forbids is an epistemically consequential distinction with no traceable ground, a false attribution of that distinction to the wrong ground, or a representation of an augmented conclusion as though it were contained in the original source.

The distinction between *truth*, *source support*, and *source attribution* is crucial. A claim may be true while its source attribution is false. A claim may also be well warranted overall through a plurality of grounds while not being warranted *by the source to which it is credited.* The title's reference to epistemic warrant is therefore not the claim that factorization is a sufficient condition for knowledge. Rather, the paper argues that source-relative warrant is a genuine component of epistemic evaluation, and that misattribution can become a warrant defeater when the alleged source is an essential ground of the subject's justification or assertion.

Conceptual and operational lineage. The Readout Condition is a standalone philosophical extraction from a broader public Readout Universe programme rather than a vocabulary assembled for this manuscript. The repository's July 2026 *Information DNA* already treated distinguishability, retention, accessibility, identity-locking, and lens discipline as distinct requirements; its Translation Protocol imposed an *identifiability gate* before an answer could be asserted, returning "structurally unanswerable from these records" rather than guessing across a null direction; its Gate Typing Law required an evidential gate to possess actual discriminating power; and by 3 August 2026 its Claim IR represented claims with explicit dependencies, tiers, and bridge classes and refused silent promotion of untranslated or bridge-dependent material (Lahtee, 2026). Public commit history timestamps these components before the present Paper Zero.[1] These artifacts do not prove the Readout Condition. They matter because they establish the independent architecture from which the present distinction-level epistemology is being abstracted.

The philosophical problem nevertheless has many near neighbours. Discrimination, relevant alternatives, and margin-for-error constraints are established themes in epistemology (Dretske, 1970; Goldman, 1976; Pritchard, 2010; Williamson, 2000, 2013; Melchior, 2021). Toulmin's argument model already distinguishes data, warrant, backing, qualifier, and rebuttal (Toulmin, 1958). Contrastivism makes the question and its alternatives constitutive of knowledge attribution (Schaffer, 2005, 2007). Manski's partial-identification programme explicitly distinguishes what data identify from what stronger assumptions identify (Manski, 2003, 2007). Blackwell comparison, statistical sufficiency, and data-processing results constrain what processing can add; philosophy of measurement distinguishes indication, model, calibration, and phenomenon; testimony and provenance research distinguish transmission, generation, assertion, and source history. The burden of this paper is therefore not to rediscover any of these components. It is to show that their conjunction still leaves a distinct object: the *typed provenance of each discriminative commitment in a claim and the way epistemic defeat propagates through that provenance structure.*

The formal development has four parts. First, exact readouts yield a fiber/factorization condition. Second, non-transitive indiscriminability requires a pointwise condition on an actual claim token rather than a global equivalence closure. Third, stochastic garbling is retained but explicitly demoted to what it can actually test: whether a downstream channel uses target-sensitive access not available through the stated source. A second, identification-based audit is required to locate assumption-driven specificity. Fourth, claim distinctions are placed in a typed provenance DAG whose dependency structure determines where defeaters and corrections propagate.

The cross-traditional route is not decorative. Dignāga–Dharmakīrti force a separation between access and conceptual uptake; Madhyamaka forces constitution-neutrality and, through the *Vigrahavyāvartanī*, presses the meta-question of how a purported means of knowledge is itself warranted; Avicenna tests whether supplementation can be classified by epistemic role rather than faculty; and Suhrawardī blocks any definition of readout that presupposes representation. Importantly, these pressures changed the framework: earlier global, representational, and faculty-typed formulations are not retained merely to make the tests pass.

The resulting thesis can be stated without inflation:

$$\boxed{\text{No epistemic discrimination without provenance.}} \tag{1}$$

The rest of the paper specifies what that maxim means, what it does not mean, and what would falsify the proposed architecture.

2. Literature Review: Classical Pressure Tests and Contemporary Collisions

2.1 Scope and comparative method

The Readout Condition addresses a problem that predates contemporary epistemology: how far may a claim legitimately distinguish among alternatives, given the way its target becomes epistemically accessible? This section develops a problem-driven comparative architecture around four classical and pre-modern pressure tests—Dignāga–Dharmakīrti, Madhyamaka, Avicenna, and Suhrawardī—while placing the closest contemporary theories alongside them.

The comparison is deliberately non-reductive. *pratyakṣa* is not identified with readout; Madhyamaka is not redescribed as contrastive semantics; Avicennian intellection is not assimilated to Bayesian inference; and *'ilm al-ḥuḍūrī* is not treated as an early theory of information channels. Each tradition is used because it destabilizes a different assumption that a general theory of epistemic access might otherwise import unnoticed. The four pressures are deliberately distinct: Dignāga–Dharmakīrti test whether access can be separated from conceptual uptake; Madhyamaka tests whether discrimination can remain neutral about intrinsic constitution; Avicenna tests the classification of supplementation; and Suhrawardī tests whether access must be representational.

These traditions constrain the neutrality of the Readout Condition rather than supplying its historical ancestry. Contemporary theories are introduced as collision tests: they determine which parts of the framework are already present in epistemology, semantics, statistics, information theory, philosophy of measurement, and provenance research.

2.2 Dignāga–Dharmakīrti: access and conceptual uptake

Dignāga's *Pramanasamuccaya* and Dharmakīrti's subsequent epistemological programme distinguish perception and inference as fundamental modes of warranted cognition. In the perceptual chapter of the *Pramanasamuccaya*, Dignāga characterizes perception through freedom from conceptual construction; Dharmakīrti develops a richer account involving particulars, causal efficacy, conceptual cognition, and exclusion (Dignaga, 1968; Dunne, 2004). These doctrines should not be simplified into a modern contrast between raw data and interpretation. Their value here is structural: what becomes available in cog-

[1] For lineage rather than evidential support: the v2 Information DNA/Translation scaffold entered the public repository in commit `1b1ca4f` (19 July 2026); Claim IR with explicit dependency fields entered in commit `a871db7` (3 August 2026). The present paper does not import the broader programme's physical or metaphysical commitments.

nition and what conceptual activity later predicates or classifies need not be the same epistemic operation.

The Readout Condition therefore does not identify *pratyakṣa* with R. It extracts a narrower pressure: In compressed form, the distinction is: *what access makes available need not exhaust what is later asserted on its basis.* If a later claim contains distinctions not attributable to the original mode of access, those distinctions require an epistemic account of how they entered.

This pressure immediately collides with contemporary discrimination-based epistemology. Goldman places reliable discriminatory mechanisms near the centre of perceptual knowledge (Goldman, 1976). Pritchard distinguishes discriminating from favouring epistemic support, weakening any simple identification of support with direct discriminatory capacity (Pritchard, 2010). Melchior develops discrimination itself as a modal epistemic capacity rather than merely a component of a theory of knowledge (Melchior, 2021). These works block any novelty claim that epistemology has neglected discrimination. The residual question is narrower: not merely whether discrimination occurs, but what licenses each discrimination asserted by a claim.

2.3 Madhyamaka: constitution-neutrality

In the *Mulamadhyamakakarika*, Nāgārjuna attacks explanations that depend on *svabhāva*, an intrinsic or independently grounded nature. Major contemporary interpretations differ over the exact reconstruction of the target, but dependence and the rejection of an ontologically self-sufficient terminus remain central (Garfield, 1995; Siderits and Katsura, 2013; Westerhoff, 2009). The Readout Condition does not derive local contrast spaces from Madhyamaka. Its narrower role is to pressure an unnoticed metaphysical assumption: epistemically relevant distinctions need not be treated as intrinsic divisions already carved into a fully specified target.

Thus the notation $R : \mathcal{X} \to Y$ must not silently turn $\mathcal{X}$ into a global inventory of independently determinate world states. $\mathcal{X}$ may instead be local to the question a claim undertakes to answer. Madhyamaka motivates **constitution-neutrality**: the framework should remain agnostic about whether relevant alternatives correspond to intrinsic joints of reality, relational structures, conventional individuations, or some combination thereof.

A different problem remains: if $\mathcal{X}$ is local, what fixes it? Here the formal resources come from contemporary work on subject matter and questions under discussion. Yablo develops subject matter through structured partitions of logical space (Yablo, 2014); Roberts models discourse around questions represented through alternative answers (Roberts, 2012). The Readout Condition uses this literature to constrain the contrast space without adopting a global possible-world ontology. The contrast space must preserve distinctions constitutive of the question the claim undertakes to settle; removing otherwise relevant alternatives is itself an epistemically significant restriction.

2.4 Avicenna: the classification of supplementation

Avicenna's psychology does not permit a simple model in which sensory input contains a complete epistemic message that intellect merely decodes. The *Kitab al-Nafs* distinguishes external and internal senses, imagination, memory, estimation, and intellectual activity; the acquisition of intelligibles also involves syllogistic reasoning, middle terms, and the Active Intellect (Avicenna, 1959; McGinnis, 2010). Specialist scholarship emphasizes the complexity of the relationship between abstraction and the Active Intellect rather than reducing one to the other (Alpina, 2014).

This makes Avicenna a pressure test for the proposed distinction between *access augmentation* and *assumption augmentation*. He is not invoked as though he already endorsed that taxonomy. Instead, his theory asks whether greater cognitive discrimination can be classified by epistemic role rather than by faculty. An additional sensory channel is not automatically "access," nor is intellectual activity automatically "assumption." The relevant question is what the addition contributes to the claim's epistemic basis.

A successful taxonomy must therefore classify supplementation functionally: does it make new target-supported distinctions available, or does it increase specificity by adding inferential commitments to what is already available? If a case cannot be assigned without distortion, the framework rather than Avicenna requires revision.

2.5 Suhrawardī: selectivity without representation

Suhrawardī provides the strongest pressure against identifying readout with representation. In *Ḥikmat al-Ishrāq*, he develops an Illuminationist account of self-awareness and presence while criticizing Peripatetic accounts in which knowledge is explained through an acquired form in the knower (al Din Yahya Suhrawardi, 1999). Kaukua's reconstruction places this development within a sustained Islamic philosophical debate over first-person awareness and presence (Kaukua, 2015).

If a readout were defined as an internal representation, the claimed generality of the Readout Condition would fail. The weaker conception required here is relational: a readout is epistemically relevant insofar as an access relation preserves or makes available distinctions that can support a claim. Representation may implement such a relation, but it is not constitutive of it. This yields a central distinction:

$$\text{Representationality} \neq \text{Selectivity}. \tag{2}$$

Even granting a maximally strong case of unmediated presence, the Readout Condition would simply become non-restrictive with respect to distinctions fully available under that relation. This is a conditional result of the framework, not a historical thesis attributed to Suhrawardī.

2.6 Nearest-neighbour collision audit

The historical pressure tests establish neutrality requirements, not novelty. The novelty question requires collision with the nearest contemporary architectures, including several that are closer than a generic provenance survey.

2.6.1 Argument warrants, contrastive knowledge, and discrimination

Toulmin's model already decomposes argument into data, claim, warrant, backing, qualifier, and rebuttal (Toulmin, 1958). It is therefore a serious predecessor for any claim about "warrant" and claim-level support. The Readout Condition does not replace that architecture. Its different object is the *discriminative capacity of the attributed source*: a Toulmin warrant can state why data support a claim without representing which target alternatives the data fail to distinguish or whether a specific claim distinction has been credited to the wrong source.

Likewise, Hintikka, Dretske, Goldman, Pritchard, Williamson, and Melchior provide powerful accounts of epistemic alternatives, information, discrimination, and indiscriminability (Hintikka, 1962; Dretske, 1970, 1981; Goldman, 1976; Pritchard, 2010; Williamson, 2000, 2013; Melchior, 2021). Schaffer's contrastivism makes an explicit contrast constitutive of knowledge attribution and later treats knowing-wh as knowing an answer to a question (Schaffer, 2005, 2007). These literatures defeat any novelty claim for discrimination or local alternatives. They do not yet yield a typed ledger saying which source, access-model, background claim, relevance restriction, or policy node is responsible for each distinction in the downstream claim.

2.6.2 Partial identification and statistical sufficiency

Manski is the closest neighbour for the assumption side of the architecture. Partial identification begins by asking what the data alone identify and then studies how identification regions shrink under additional assumptions (Manski, 2003, 2007). This paper adopts that lesson directly: a stochastic garbling relation cannot reveal an undeclared prior or model assumption when the downstream rule consumes only the observed readout. An assumption audit therefore requires an identification ladder, not merely an information-ordering test. The residual contribution is to integrate this assumption ladder with source attribution, disclosure, readout models, iterated provenance, and defeater routing at the level of claim distinctions.

Classical Fisher–Neyman sufficiency also blocks any suggestion that preserving all information relevant to a target parameter is new (Fisher, 1922; Neyman, 1935). A sufficient statistic is a compression that loses no information about the parameter under a specified statistical model. The Readout Condition is neither a rival definition of sufficiency nor a claim that its deterministic factorization theorem is new. Its target is broader and differently typed: what an epistemic source may be credited with across deterministic, relational, stochastic, testimonial, model-mediated, and institutional routes.

2.6.3 Blackwell, Dretske, measurement, and robust access

Blackwell's comparison of experiments and the data-processing inequality formalize monotonicity under post-processing (Blackwell, 1951, 1953; Cover and Thomas, 2006). Dretske's information theory stresses that what a signal carries is relative to channel conditions (Dretske, 1981). Suppes, Bogen and Woodward, Tal, Giere, and Massimi distinguish data, phenomena, measurement models, calibration, and perspective (Suppes, 1962; Bogen and Woodward, 1988; Tal, 2017; Giere, 2006; Massimi, 2022). Hacking and Wimsatt make independent routes and robustness epistemically important (Hacking, 1983; Wimsatt, 1981). These are not competitors to be displaced; they identify distinct typed nodes in the provenance graph. The Readout Condition's claim is that those nodes must not be silently collapsed when downstream specificity is represented.

2.6.4 Testimony and assertion provenance

Lackey's rejection of simple transmission shows that testimonial epistemic status can be generated or transformed rather than merely passed intact from speaker to hearer (Lackey, 2008). Nanopublications and related provenance infrastructures already package assertions with contextual provenance (Groth et al., 2010; Chhetri et al., 2025). These developments defeat any claim that assertion-level provenance or dependency is new. The residual distinction is finer: documentary or assertion provenance can be correct while the *discriminative dependency* of an assertion remains misdescribed. A claim may cite the correct dataset and still represent a prior-induced distinction as though the dataset itself distinguished it.

The novelty claim is therefore a conjunction claim. No individual mathematical or philosophical component is presented as unprecedented. What is proposed is a domain-neutral architecture in which (i) the audit unit is an individual claim distinction, (ii) source attribution has an explicit licensing condition, (iii) assumptions are located by an identification ladder, (iv) provenance is a typed dependency DAG, and (v) source-specific defeaters propagate through that DAG. The public Readout Universe lineage supplies an operational precursor for this conjunction—identifiability gates, bridge classes, tier downgrades, and explicit claim dependencies—but the present paper is the first place in that programme where those components are reconstructed as a standalone epistemic condition.

3. Conceptual Method

3.1 Methodological status

This is a work of conceptual and formal philosophy, not an empirical study. The method combines problem reconstruction, formal reconstruction, cross-traditional pressure testing, nearest-neighbour collision testing, and self-application to the paper's own claims. The last component matters: a provenance theory that exempts its own models, citations, or formal bridges from audit would reproduce the very silent lift it criticizes.

Table 1: Nearest-neighbour collision audit. The contribution defended here is the conjunction, not the individual components.

Neighbour	**Already provides**	**What the Readout Condition concedes**	**Residual task**
Toulmin	Data–claim warrants, backing, rebuttal	Claim-level licensing is not new	Test source discrimination and misattribution
Schaffer / QUD	Explicit contrast structure	Local alternatives are not new	Track provenance of each resolved contrast
Goldman / Pritchard / Melchior / Williamson	Discrimination, support, non-transitivity	Discrimination is not new	Pointwise source licensing plus provenance
Manski	Data-only vs assumption-dependent identification	Assumption burden is not new	Type the weakest augmentation layer for each distinction
Blackwell / DPI / sufficiency	Informativeness and lossless processing	Monotonicity/factorization are not new	Separate access audit from assumption audit
Dretske / measurement	Channel conditions, model mediation	Source models are not transparent	Distinguish K^* from $\widetilde{K}$ and route defeat
Lackey / robustness	Generation, transmission, independent routes	Epistemic routes may transform status	Represent branching and recursive access
Nanopublications / provenance	Assertion and source lineage	Provenance simpliciter is not new	Distinction-level dependency and defeater propagation

3.2 Units and roles

Let Q determine a local contrast space $\mathcal{X}$. Let R denote a deterministic readout, I_R a relational access structure, or K^* a stochastic access experiment. Let Φ represent a categorical claim structure. The unit of audit is not only the whole proposition but a **claim distinction**. For an actual alternative x_0, define

$$D_\Phi(x_0) = \{d_{0x'} = (x_0, x') : \Phi(x') \neq \Phi(x_0)\}. \tag{3}$$

Each $d_{0x'}$ is one contrast the actual claim token resolves against a rival alternative.

A provenance analysis assigns each such distinction a directed acyclic dependency graph

$$\mathcal{P}(d) = \big(V_d, E_d, \tau_d\big), \tag{4}$$

where the typing map τ_d distinguishes at least: *access source*, *access model*, *background-evidence claim*, *contrast/relevance operation*, *inferential commitment*, and *decision policy*. A single object may occupy different types in different audits; the types are epistemic roles, not ontological kinds.

This formal move is directly continuous with the public programme's earlier Claim IR, which already represented claims with dependencies, tiers, and bridge classes and automatically prevented untranslated or bridge-dependent material from being silently promoted. The present graph generalizes that operational discipline from whole claims to discriminative commitments.

3.3 Internal reconstruction and pressure-driven revision

Historical comparison proceeds by *internal reconstruction* $\to$ *controlled abstraction* $\to$ *pressure test.* A pressure test counts only if it can force a change. In fact, the present framework records four such changes: a global contrast-space formulation was replaced by a local one under Madhyamaka/QUD pressure; a representation-defined readout was replaced by a selectivity role under Suhrawardī pressure; a faculty-based supplementation taxonomy was replaced by role typing under Avicennian pressure; and the original global relational condition was replaced by a pointwise claim-token condition after the Williamson-style non-transitivity objection. These are theory revisions, not retroactive declarations that every tradition "passes."

3.4 Collision criteria

For each neighbour the paper asks: does it analyze the same object, impose the same constraint, type the same epistemic sources, locate assumption dependence, and determine how defeat propagates? Novelty survives only at the residual conjunction after the strongest relevant results are granted.

3.5 Failure and revision conditions

The architecture fails, rather than merely losing elegance, if (i) a source can be correctly credited with a distinction it does not locally discriminate without any other epistemic ground; (ii) the provenance graph cannot represent an important supplementation case without ad hoc typing; (iii) two genuinely different dependency structures always generate identical patterns of rational defeat and correction; or (iv) an existing theory already supplies the same distinction-level licensing, identification, provenance, and defeat architecture at comparable generality. Application-level opacity is different: when K^*, $\widetilde{K}$, or the historical dependency graph cannot be recovered, the correct output is provenance indeterminacy, not forced classification.

4. The Readout Condition

4.1 Local contrasts and claim discrimination

Let Q be a question or epistemic task. Its relevant alternatives determine a local contrast space $\mathcal{X}_Q$, abbreviated $\mathcal{X}$. The local qualification matters. A claim is never audited against every imaginable difference in reality; it is audited against the distinctions it undertakes to settle. At the same time, the claimant cannot freely shrink $\mathcal{X}$ after inspecting the evidence. A relevance restriction that removes otherwise constitutive alternatives is itself an epistemically significant commitment.

A claim structure Φ can be represented, in the exact categorical case, as a function

$$\Phi : \mathcal{X} \to Z, \tag{5}$$

where Z indexes the distinctions made by the claim. Two alternatives $x, x' \in \mathcal{X}$ are claim-equivalent when $\Phi(x) = \Phi(x')$ and claim-distinguished otherwise.

A readout R specifies what the relevant access relation preserves. The exact deterministic case is the simplest:

$$R : \mathcal{X} \to Y. \tag{6}$$

Nothing in this notation implies that the target is metaphysically discrete or that the readout is a representation. The functions describe the contrastive role played by the access structure relative to the current question.

4.2 Exact functional case

Definition 1 (Readout-admissible discrimination)**.** *A categorical claim structure $\Phi : \mathcal{X} \to Z$ is* readout-admissible relative to $R : \mathcal{X} \to Y$ *when it is constant on every fiber of R:*

$$R(x) = R(x') \Rightarrow \Phi(x) = \Phi(x'). \tag{7}$$

Proposition 1 (Factorization)**.** *For functions $R : \mathcal{X} \to Y$ and $\Phi : \mathcal{X} \to Z$, the following are equivalent:*

(i) *Φ is constant on every fiber of R;*
(ii) $\ker R \subseteq \ker \Phi$*;*
(iii) *there exists a function $\bar{g} : \mathrm{im}(R) \to Z$ such that $\Phi = \bar{g} \circ R$.*

Proof. If $\Phi = \bar{g} \circ R$, then $R(x) = R(x')$ implies $\Phi(x) = \Phi(x')$, so (iii)$\Rightarrow$(i). Statement (i) is exactly the inclusion of the equivalence relation induced by R in that induced by Φ, giving (i)$\Leftrightarrow$(ii). For (i)$\Rightarrow$(iii), define $\bar{g}(y) = \Phi(x)$ for any x such that $R(x) = y$. Fiber constancy makes this well-defined on $\mathrm{im}(R)$. □

The proposition is elementary mathematics. The philosophical claim is not that factorization is new, but that factorization marks a boundary on *source attribution*. If $R(x) = R(x')$ while $\Phi(x) \neq \Phi(x')$, the distinction in Φ did not come from R alone. A broader epistemic case may still justify the claim, but the additional discrimination requires another source.

This yields the exact Readout Condition:

Principle 4 (Exact Readout Condition)**.** *If a categorical distinction in Φ is licensed by R alone, then Φ must be readout-admissible relative to R.*

The direction is intentionally one-way. Factorization is sufficient for formal admissibility, not for epistemic warrant as a whole.

4.3 Relational readouts: pointwise licensing

Many access relations cannot be represented adequately by partitions. Indiscriminability may be reflexive and symmetric without being transitive (Williamson, 2013). Let $I_R \subseteq \mathcal{X} \times \mathcal{X}$ and define the local indiscriminability neighbourhood of an actual alternative x_0 as

$$N_R(x_0) = \{x' \in \mathcal{X} : (x_0, x') \in I_R\}. \tag{8}$$

Let the actual claim token have value $z_0 = \Phi(x_0)$.

Principle 5 (Pointwise Relational Readout Condition)**.** *The claim token $z_0 = \Phi(x_0)$ is licensed by relational access R alone only if*

$$\boxed{N_R(x_0) \subseteq \Phi^{-1}(z_0).} \tag{9}$$

Equivalently, the claim token may not exclude any alternative that remains locally indiscriminable from the actual case under the attributed source.

The quantifier is deliberately pointwise. A global requirement $I_R \subseteq I_\Phi$ would indeed force every equivalence relation containing I_R to contain its transitive closure and would collapse a Sorites chain. Equation (9) does not iterate licenses from one centre to another: licensing at x_0 constrains $N_R(x_0)$ but does not imply licensing at every $x' \in N_R(x_0)$. The local geometry therefore remains visible.

Proposition 2 (Reduction to the deterministic case)**.** *If I_R is induced by a deterministic readout R through $xI_Rx' \iff R(x) = R(x')$, then requiring (9) for every $x_0 \in \mathcal{X}$ is equivalent to fiber constancy and hence to factorization $\Phi = \bar{g} \circ R$ on $\mathrm{im}(R)$.*

This is the appropriate sense in which the relational condition extends the exact case: not by replacing a non-transitive relation with an equivalence closure, but by changing the object of licensing from a global claim function to an actual claim token.

4.4 Stochastic readouts: access audit, not assumption audit

Let the actual source relation be a Markov kernel

$$K^* : Y \mid \mathcal{X}, \tag{10}$$

and a stochastic downstream output be a channel $C : Z \mid \mathcal{X}$. For the finite case, Blackwell comparison gives the following access test; for more general experiment spaces the relevant extensions belong to the Le Cam–Torgersen theory of statistical experiments (Blackwell, 1951, 1953; Cam, 1964; Torgersen, 1991).

Principle 6 (Stochastic access admissibility)**.** *A downstream channel uses no target-sensitive access beyond K^* only if there exists a Markov kernel $G : Z \mid Y$ such that*

$$\boxed{C = G \circ K^*.} \tag{11}$$

The wording is intentionally precise. Equation (11) is an *access audit*, not a complete provenance audit. Any rule that consumes only y—including one that uses an undeclared prior, model class, or loss function—can be absorbed into G. Garbling therefore detects an undeclared target-sensitive route that bypasses the stated source; it does *not* by itself detect silent assumption augmentation.

Treating it as a full source-only warrant condition would be a category error.

Practical inference also distinguishes the actual access process from its model:

$$\boxed{K^* \neq \widetilde{K}.} \tag{12}$$

Likelihood ratios are ordinarily computed under $\widetilde{K}$, not read directly from K^*:

$$\Lambda_{\widetilde{K}}(y; x, x') = \frac{\widetilde{K}(y \mid x)}{\widetilde{K}(y \mid x')}. \tag{13}$$

In a Bayesian representation,

$$\log \frac{P(x \mid y)}{P(x' \mid y)} = \log \frac{P(x)}{P(x')} + \log \Lambda_{\widetilde{K}}(y; x, x'). \tag{14}$$

The decomposition already shows that source model and prior occupy different provenance roles. The next section supplies the missing assumption audit.

5. Augmentation and Epistemic Provenance

5.1 Typed augmentation grammar

The phrase "assumption augmentation" is too coarse if it treats every downstream addition as the same operation. The provenance graph therefore distinguishes four signatures.

Definition 2 (Access augmentation)**.** *A new target-sensitive route is added to the epistemic basis, for example a second measurement, testimony, retrieved record, sensor, or retained upstream observation. Formally, the observation structure expands, e.g.* $Y_1 \mapsto (Y_1, Y_2)$*.*

Definition 3 (Contrast or relevance operation)**.** *The contrast domain changes,* $\mathcal{X} \mapsto \mathcal{X}' \subseteq \mathcal{X}$*. Such restriction can be legitimate when fixed by the question or independently justified. A restriction selected after inspecting the readout requires its own provenance because it reuses the observed result in defining what alternatives count.*

Definition 4 (Inferential commitment)**.** *A model, prior, theory, bridge principle, calibration assumption, or other rule changes what can be inferred from an existing access basis without constituting new current access to the target.*

Definition 5 (Decision-policy augmentation)**.** *A loss function, threshold, utility, or institutional rule maps an epistemic state into an action or classification. It can change the decision while leaving the evidential discrimination unchanged.*

Background evidence is represented recursively rather than forced into a "mere assumption" box. A population prior may be an imported commitment relative to today's patient test while itself being the conclusion of an earlier evidence graph. This recursion is essential to the architecture inherited from the Readout Universe claim/tier discipline: a bridge-dependent claim may be usable, but the bridge never disappears from its lineage.

5.2 Identification ladder: where a distinction first becomes available

Manski's partial-identification programme supplies the right mathematical resource for the assumption side of the audit (Manski, 2003, 2007). Let $\mathcal{A}_0$ denote the weakest declared commitment set compatible with the access model, and let

$$\mathcal{A}_0 \subseteq \mathcal{A}_1 \subseteq \cdots \subseteq \mathcal{A}_m \tag{15}$$

represent increasingly strong augmentation layers. For each layer let Γ_j be the set of target alternatives not excluded by the access model plus commitments through $\mathcal{A}_j$.

For an actual claim token $z_0 = \Phi(x_0)$, define the first-identification level

$$\ell(d_{0x'}) = \min\{j : x' \notin \Gamma_j\}, \tag{16}$$

whenever the rival x' is excluded at some layer. The provenance of that discrimination is not "the data" simpliciter; it includes the weakest augmentation layer at which the rival becomes excludable. If x' is already excluded at $j = 0$, the discrimination is access-model licensed at baseline. If exclusion begins only after a prior, structural model, or relevance restriction enters, the distinction is assumption-dependent. If no declared layer excludes it, the distinction remains unidentified.

Equation (16) is deliberately an audit definition rather than a new theorem in partial-identification theory. Its contribution is to attach assumption strength to individual claim distinctions and then connect those distinctions to attribution and defeat.

5.3 The E–A–D form of the Readout Condition

Let $\mathcal{P}(d)$ be the typed provenance DAG for distinction d. The mature condition is:

Principle 7 (E: provenance existence)**.** *For every epistemically load-bearing claim distinction* d*,* $\mathcal{P}(d)$ *is nonempty or the claim is explicitly marked provenance-indeterminate.*

Principle 8 (A: provenance attribution)**.** *If the public or doxastic attribution map* $\widehat{\lambda}(d)$ *credits distinction* d *to source* S*, then* S *must pass the appropriate licensing test for* d*: fiber discrimination in the exact case, pointwise neighbourhood exclusion in the relational case, or access admissibility/identification in the stochastic case.*

Principle 9 (D: provenance disclosure)**.** *Every non-source node on an essential dependency path to* d *must remain declared or recoverable at the level relevant to the epistemic practice. Omitting such a node while representing* d *as source-generated is a silent lift.*

The Readout Condition is $E \wedge A \wedge D$. This separation closes an ambiguity in weaker formulations: the mere existence of a valid augmentation does not rescue a false attribution to the original source.

5.4 Epistemic overreach and silent lift

Definition 6 (Epistemic overreach). *A claim-level distinction exhibits epistemic overreach when it lacks an adequate provenance path, is attributed to a node that does not license it, or depends on an essential augmentation that is concealed in the relevant epistemic representation.*

Definition 7 (Silent lift). *Let $\lambda(d)$ denote the actual essential dependency set of distinction d and $\widehat{\lambda}(d)$ the dependency set represented to oneself or others. A silent lift occurs when an essential node $v \in \lambda(d)$ is omitted or a source S is credited with d despite failing the corresponding licensing condition.*

This is not an anti-inference principle. A model-dependent or prior-dependent distinction can be fully warranted when the dependency is genuine and the supporting background claims are themselves adequate. The target is false provenance, not epistemic sophistication.

5.5 Defeater routing on the provenance DAG

The payoff of distinction-level provenance is not documentary neatness but revision structure. For a distinction d, let $\Pi(d)$ be the set of currently adequate licensing paths in $\mathcal{P}(d)$, each running from epistemic leaves to d. Define the essential dependency set

$$\mathrm{Ess}(d) = \bigcap_{p \in \Pi(d)} V(p). \tag{17}$$

Thus a node is essential exactly when every currently adequate licensing path for d passes through it.

Let Δ_v be a node-specific defeater that defeats provenance node v while leaving all other nodes and edges initially unchanged.

Proposition 3 (Defeater Routing). *After Δ_v, the surviving licensing paths for distinction d are*

$$\boxed{\Pi_{\Delta_v}(d) = \{p \in \Pi(d) : v \notin V(p)\}.} \tag{18}$$

Consequently, d loses provenance support under the node-specific defeat iff $\Pi_{\Delta_v}(d) = \varnothing$, equivalently iff $v \in \mathrm{Ess}(d)$.

Argument. A path that contains the defeated node cannot remain an adequate licensing path; a path that does not contain it is untouched by a defeater whose stipulated target is v alone. Equation (18) follows by path deletion. The distinction loses all current provenance support exactly when no licensing path survives, which is equivalent to v belonging to every path in $\Pi(d)$, i.e. $v \in \mathrm{Ess}(d)$. The epistemic interpretation is substantive even though the graph step is elementary: equal propositional content can survive different challenges because its licensing paths differ. □

Corollary 1 (Misrouted defeat under silent lift). *Let* $\mathrm{Ess}(d)$ *be the actual essential dependency set and* $\widehat{\mathrm{Ess}}(d)$ *the represented one. If the symmetric difference is nonempty, then for any epistemically possible node-specific defeater aimed at a node in that difference, the represented vulnerability of d differs from its actual dependency vulnerability. Silent lift can therefore generate under-revision when a hidden essential node is defeated, and over-revision when an innocent represented node is treated as essential.*

This result formalizes a discipline that the public Readout Universe implemented operationally before this paper: bridge-dependent claims carry dependency and tier information; untranslated terms are forced to Open rather than silently promoted; the identifiability gate blocks answers outside the record's row space; and a gate with no discriminating controls is classified as convention rather than evidence. Those rules are not proofs of Proposition 3, but they are working instances of the same dependency-sensitive norm.

5.6 From provenance error to epistemic warrant

Why does false attribution matter to warrant rather than merely to reporting? Consider a subject who accepts distinction d partly because she believes the test, witness, or dataset S itself discriminates the relevant alternatives. If correcting that meta-belief—learning that S did not discriminate them and that the result depended on an undeclared bridge—would rationally require withdrawal or suspension of d absent another ground, then the false attribution is an *essential ground* in her justificatory structure. Debates over inference from falsehood show that false premises need not always destroy knowledge, but they make the essentiality of the false ground the relevant issue (Warfield, 2005; Klein, 2008). The Readout Condition therefore identifies a class of genuine warrant failures: cases in which a false provenance belief is load-bearing for justification.

The framework also has an assertoric face. A privately warranted belief may rely on a model or prior without defect, yet asserting that "the test showed d" can misrepresent the epistemic basis when the test alone did not. Testimonial epistemology already distinguishes the epistemic status of speakers, statements, and hearers and rejects simple transmission pictures (Lackey, 2008). The Readout Condition adds a provenance norm on how the discriminative basis of an assertion is represented. Doxastic warrant and assertoric disclosure are related but not identical obligations.

6. Iterated Readout Chains

6.1 Deterministic composition

Readouts frequently occur in chains. A record may be produced by an instrument, transformed by software, summarized in a database, processed by a model, and interpreted by a human. Let

$$\mathcal{X} \xrightarrow{R_1} Y_1 \xrightarrow{R_2} Y_2 \to \cdots \xrightarrow{R_n} Y_n. \tag{19}$$

If $R_1(x) = R_1(x')$, then every deterministic downstream composition also maps x and x' to the same result. More generally, if an upstream stage has already collapsed a pair, post-processing alone cannot recreate that pairwise distinction.

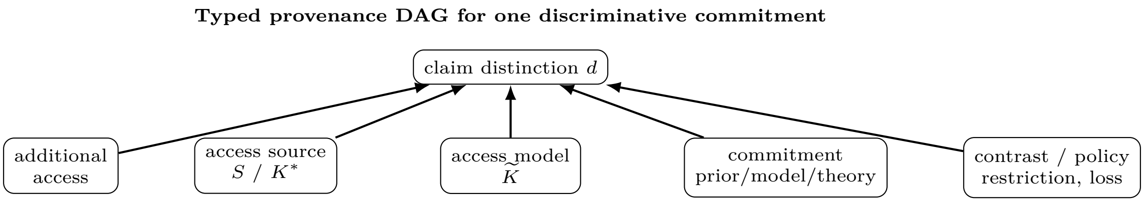

Figure 1: A claim distinction can depend on several differently typed nodes. The Readout Condition audits existence, attribution, and disclosure; Proposition 3 uses the same graph to route defeat and correction.

Corollary 2 (Upstream monotonicity)**.** *For a deterministic chain $R_{1:n} = R_n \circ \cdots \circ R_1$,*

$$\ker R_1 \subseteq \ker R_{1:n}. \tag{20}$$

At any stage j, distinctions collapsed by $R_{1:j}$ remain collapsed under later deterministic processing unless additional access enters the chain.

This is the precise form of the claim that target-supported discrimination cannot increase downstream merely through processing. The formulation avoids talk of a single "coarsest link," which is not generally well-defined across heterogeneous stages.

6.2 Stochastic composition

For stochastic chains, let the actual upstream experiment be $K_1^* : Y_1 \mid \mathcal{X}$ and let later stages be Markov kernels. Whenever $K_{j+1}^* = G_j \circ K_j^*$, each downstream experiment is a garbling of the preceding one and is therefore weakly less informative in the Blackwell order. Standard data-processing inequalities provide metric-specific consequences, for example

$$I(X; Y_{j+1}) \leq I(X; Y_j) \tag{21}$$

for mutual information in a Markov chain (Cover and Thomas, 2006). The Readout Condition does not depend on mutual information as the unique measure of distinguishability; the Blackwell relation is used because it expresses the more general experiment-level ordering.

A downstream channel C passes the access-admissibility test relative to K_j^* only if $C = G \circ K_j^*$ for some post-processing kernel G. This establishes that no extra target-sensitive access route is required; it does not establish that the downstream specificity is assumption-free. The identification ladder and provenance DAG perform that second audit. Additional access changes the architecture by forming a joint experiment with a genuinely new evidential path.

6.3 Calibration and the model of the readout

A claimant rarely knows the actual access structure directly. Let R^* or K^* denote the source relation under analysis and let $\widetilde{R}$ or $\widetilde{K}$ denote the claimant's model of it. Tal's analysis of calibration is important here because calibration constructs and tests models that connect indications with measurement outcomes (Tal, 2017). The Readout Condition therefore distinguishes

$$\boxed{R^* \neq \widetilde{R}, \qquad K^* \neq \widetilde{K}} \tag{22}$$

as epistemic roles even when the model is excellent.

This distinction prevents a subtle silent lift in stochastic inference. A likelihood ratio computed from $\widetilde{K}$ is evidence *under a model of the source*; it is not literally read off from K^*. Calibration evidence can license the use of $\widetilde{K}$, but its contribution must remain visible.

At least three failure modes follow. In an **access failure**, R^* or K^* does not preserve the distinction required by the claim. In an **access-model failure**, $\widetilde{R}$ or $\widetilde{K}$ mischaracterizes what the source preserves. In an **attribution failure**, a distinction introduced by another source or commitment is credited to the original source. These failures require different remedies: additional or improved access, recalibration/model revision, or corrected provenance attribution.

7. Classical Pressure Tests Revisited

A pressure test earns methodological force only if it can change the theory. Table 2 records the revisions produced by the four traditions rather than simply declaring that the final framework "passes."

7.1 Dignāga–Dharmakīrti: access, apoha, and perceptual judgment

The relevant pressure is sharper than a generic perception/inference contrast. Dignāga's and Dharmakīrti's accounts of conceptual exclusion (*apoha*) and later perceptual judgment make it impossible to assume that every discrimination present in a conceptual claim was already present in the same form in non-conceptual perception (Dignaga, 1968; Dunne, 2004; Arnold, 2005; Kellner, 2004). The Readout Condition therefore places conceptual uptake downstream of access and demands provenance for distinctions introduced by classificatory rules. It does not identify *apoha* with the Readout Condition; it accepts the historical challenge that discrimination can be generated by conceptual organization rather than simply copied from perception.

7.2 Madhyamaka and the warrant of the means of access

The *Mūlamadhyamakakārikā* blocks an unargued move from epistemic contrasts to intrinsic target constitution (Garfield, 1995; Siderits and Katsura, 2013; Westerhoff, 2009). The *Vigrahavyāvartanī* adds a meta-epistemic pressure: if a means of knowledge is invoked to establish an object, the status of that means cannot simply be exempted from scrutiny (Westerhoff, 2010). The parallel

Table 2: Pressure-driven revisions of the Readout Condition.

Pressure	Earlier vulnerable formulation	Failure	Revision retained here
Dignāga–Dharmakīrti	downstream classification folded into “what perception gives”	conceptual uptake inherits source status automatically	access/readout and claim uptake are separate nodes
Madhyamaka	global state space read as intrinsic constitution	formal contrast structure becomes covert metaphysics	local $\mathcal{X}_Q$ plus constitution-neutrality
Avicenna	sensory faculty = access; intellect = assumption	faculty names substitute for epistemic roles	role-typed augmentation with provenance indeterminacy
Suhrawardī	readout defined as internal representation	knowledge by presence is excluded by definition	readout = selectivity/access role; presence itself is not forced into a representation model
Williamson-style objection	global $I_R \subseteq I_\Phi$	transitive closure erases non-transitive geometry	pointwise claim-token condition $N_R(x_0) \subseteq \Phi^{-1}(z_0)$

here is controlled rather than identificatory. The Readout Condition's distinction $K^* \neq \widetilde{K}$ makes the model or calibration of access itself a claim with provenance; a failed access-model audit is not self-undermining, but a higher-level readout failure to be diagnosed under the same rules.

7.3 Avicenna: the *wahm* test

Avicenna's estimative faculty (*wahm*) is a concrete stress test because the sheep's apprehension of the wolf's hostility is not straightforwardly one more sensible quality (Avicenna, 1959; McGinnis, 2010; Alpina, 2014). The Readout Condition does not decide the historical psychology by fiat. It supplies a criterion. If the additional faculty constitutes a target-sensitive route whose output varies with the relevant *maʿnā* while the previous sensory readout is held fixed, it plays an access-augmentation role. If the hostility classification is generated from the existing readout by a standing rule or learned commitment, it plays an inferential-commitment role. If the historical evidence does not resolve which dependence obtains, the audit returns provenance indeterminacy. The taxonomy therefore answers the case conditionally by role rather than hiding the difficulty under a faculty label.

7.4 Suhrawardī: presence before claim

Knowledge by presence should not be forced into a contrastive channel when the doctrine itself denies representational mediation. The correct boundary is therefore stronger than saying that the Readout Condition becomes “non-restrictive.” Presence as such can be *pre-claim* and hence outside the condition's immediate domain. The Readout Condition applies when a categorical or otherwise discriminative claim is made on the basis of presence—for example, when “I exist” or a further articulated judgment excludes alternatives. At that point the audit asks what distinctions the presence relation licenses and what conceptual articulation adds. This preserves representation-independence without pretending to formalize the metaphysics of presence itself (al Din Yahya Suhrawardi, 1999; Kaukua, 2015; Yazdi, 1992).

8. Applications and Portability

Portability is demonstrated here by running the audit to verdict rather than merely listing domains.

8.1 Worked audit I: a positive diagnostic test

Suppose a test has sensitivity 0.90, specificity 0.91, and is used in a population with prevalence 0.01. For a positive result $y = +$, Bayes' rule gives

$$P(D \mid +) = \frac{0.90(0.01)}{0.90(0.01) + 0.09(0.99)} \approx 0.0917. \tag{23}$$

The immediate source contribution is the positive test result together with an access model $\widetilde{K}$ specifying sensitivity and specificity. The posterior discrimination between disease and no disease additionally depends on the prevalence prior. A decision to treat may depend further on a loss function or threshold.

The sentence “the test shows that the patient has the disease” therefore compresses at least three typed nodes:

$$(y, \widetilde{K}) + \pi + L \longrightarrow \text{belief/action}. \tag{24}$$

With the stated numbers the positive predictive value is only about 9.2%. The Readout Condition does not say that using the prior is illegitimate; it says that the posterior specificity cannot be attributed to the positive readout alone. If prevalence changes while sensitivity and specificity remain fixed, the claim can change without any change in the individual test result. That counterfactual identifies the prior as load-bearing. A calibration defeater targets $\widetilde{K}$; a prevalence defeater targets π; a new imaging result constitutes additional access; a threshold objection targets L. Proposition 3 therefore yields different correction routes for superficially similar diagnostic disagreement.

8.2 Worked audit II: AI-assisted inference and retained-record contamination

Let K^* represent current case-visible evidence and let a fixed trained model with training corpus D implement a downstream rule G_D. If D is fixed independently of the current case x, then

$$C_D = G_D \circ K^* \tag{25}$$

can pass the stochastic access-admissibility test even though training-derived structure materially shapes the output. This is exactly why garbling cannot serve as an assumption audit: fixed background structure can be absorbed into G_D.

Proposition 4 (Retained-record route). *If a supposedly fixed background corpus contains a current-case-specific* *retained record $D(x)$ that varies with the target alternative and is available to the model through a route not contained in the stated current source K^*, then the effective architecture contains additional access. Relative to K^* alone the output need not admit a representation $C = G \circ K^*$ with G independent of that retained record.*

This separates two phenomena often conflated under "model knowledge." Ordinary training-derived structure is a background commitment/provenance node relative to the current episode. Memorization, contamination, retrieval, or tool access that carries current-case-specific information is an additional access route. The same final answer can therefore have different epistemic vulnerability depending on which route generated its specificity.

A complete AI audit should record at least: current evidence, retrieved/tool evidence, retained-record routes, model/calibration assumptions, prompt/system constraints, and decision policy. Calling all of this "the AI's answer" erases the dependency structure that determines what can be defeated and repaired.

8.3 Portability claim

The same typed questions can be asked of testimony, measurement, historical inference, peer review, and institutional classification. Portability here means invariance of the audit questions and dependency types, not empirical validation of identical mechanisms across domains.

9. Discussion

9.1 What is actually new

The paper does not claim novelty for factorization, statistical sufficiency, Blackwell garbling, partial identification, contrastive knowledge, argument warrants, measurement-model dependence, testimony, robustness, or provenance. The contribution is the conjunction:

$$\boxed{\begin{array}{c}\text{distinction-token audit} + \text{source licensing}\\ + \text{ identification layer} + \text{typed provenance DAG}\\ + \text{ E–A–D norms} + \text{defeater routing}\end{array}} \tag{26}$$

This is not a verbal restatement of any single neighbour. Toulmin does not provide a target-discrimination test; Manski does not provide source-attribution/disclosure norms or defeater routing; Blackwell cannot expose a hidden prior inside G; nanopublications do not determine which contrast within an assertion depends on which epistemic node. The Readout Condition integrates these partial structures around a single audit object: the discriminative commitments of a claim.

9.2 Why this is not tautological bookkeeping

"A source cannot support what it does not distinguish" sounds trivial until a real inference contains models, priors, restrictions, retained records, thresholds, and institutional policies. The non-trivial content lies in preserving the types through composition and deriving revision consequences from them. Proposition 3 is the key normative payoff: two identical propositions can rationally react differently to the same challenge because their provenance graphs differ. A provenance error therefore changes defeater sensitivity, not merely metadata.

9.3 Reliabilism and the meta-level objection

A reliabilist may object that a reliable process can yield warrant regardless of whether the subject can articulate its provenance. The Readout Condition need not deny this. The distinction is between *having* a reliable route and *crediting a specific source with a discrimination*. If the process is reliable because of hidden model structure, background data, or another route, then the overall belief may be warranted while the source attribution is false. Where that attribution is itself load-bearing for justification, explanation, or assertion, the false provenance becomes epistemically consequential. The framework therefore complements rather than replaces process reliabilism.

The audit also applies to itself. Any verdict about K^* is normally mediated by $\widetilde{K}$; a mistaken access model produces an access-model failure, not a paradox that exempts meta-claims from the condition. The public Readout Universe's own discipline—tier downgrades, explicit bridge classes, identifiability gates, and gate controls—is best understood as a self-application protocol: meta-level readouts remain readouts rather than being promoted to truth by their supervisory role.

9.4 Epistemic and assertoric faces

The condition has a doxastic face and an assertoric face. Doxastically, a distinction is vulnerable when its actual dependency graph contains a defeated or falsely represented essential ground. Assertorically, a speaker can misrepresent a well-supported conclusion by describing an assumption- or model-dependent discrimination as "what the data show." The first concerns warrant and revision; the second concerns epistemic representation to others. Keeping them separate prevents "undisclosed" from doing ambiguous work.

10. Limitations and Open Problems

The elevated architecture creates sharper, not weaker, falsifiers.

First, the pointwise relational condition is categorical. A full theory of graded local discrimination may require tolerance structures, similarity orderings, or margin functions. What is fixed here is the quantifier structure needed to avoid transitive collapse.

Second, the identification ladder requires an explicit ordering or nesting of augmentation sets. In complex scientific practice, assumptions may be partially ordered rather than linearly nested. The natural extension is a provenance lattice in which a distinction's first-identification set may contain several incomparable minimal bundles.

Third, provenance DAGs can contain causal and epistemic dependencies that are difficult to distinguish empirically. The graph is therefore a normative dependency representation, not automatically a discovered causal graph. When the dependency type cannot be established, the honest state is provenance indeterminacy.

Fourth, the stochastic access condition is stated transparently as an access audit. Its inability to detect fixed background assumptions is not a residual defect but the reason for the identification layer. A counterexample would require a target-sensitive route not represented in K^* that nevertheless always passes the access audit without being absorbable into the declared architecture.

Fifth, the warrant bridge is intentionally conditional on essentiality. False provenance need not destroy every possible form of externalist warrant. The stronger claim defended is that false source attribution is a genuine defeater whenever that attribution is load-bearing for the subject's justification, for an assertion's represented basis, or for downstream correction.

Finally, novelty is a falsifiable conjunction claim. Finding prior work that already combines distinction-token source licensing, assumption-identification levels, typed provenance dependencies, attribution/disclosure norms, and defeater routing at comparable domain neutrality would defeat the originality claim. The paper does not pre-emptively narrow itself merely because adjacent components have predecessors; it states exactly what conjunction a challenger would need to match.

11. Conclusion

The Readout Condition began as a simple question—what does a source actually distinguish?—but the adversarial development of the framework shows why a one-line factorization principle is not enough. A mature audit must distinguish the existence of epistemic grounds from their attribution, and attribution from disclosure. It must distinguish actual access K^* from the model $\widetilde{K}$ by which that access is interpreted. It must prevent non-transitive relational access from being destroyed by a global closure. It must recognize that Blackwell garbling audits additional access but is blind to fixed assumptions inside post-processing. And it must say why provenance matters after a claim has already been formed.

The resulting architecture answers those demands. Exact source attribution is constrained by fiber factorization. Relational claim tokens are constrained pointwise by their local indiscriminability neighbourhoods. Stochastic access is audited by experiment ordering, while assumption dependence is located through an identification ladder. Individual claim distinctions carry typed provenance DAGs linking access, models of access, background evidence, relevance operations, inferential commitments, and policies. The E–A–D form of the condition then requires provenance existence, correct source attribution, and recoverable essential augmentation.

The strongest result is normative rather than algebraic: provenance determines the topology of rational revision. Under Proposition 3, a defeater propagates along the distinctions that depend on its target node; independent paths can preserve content; and silent lift misroutes defeat and correction. This is why equal propositional content can have unequal epistemic vulnerability and why provenance is not metadata appended after warrant has been settled.

The four classical pressure tests remain part of the architecture because they forced real revisions: local rather than globally metaphysical contrasts, selectivity rather than representationality, role-typed rather than faculty-typed supplementation, and a boundary between pre-claim presence and discriminative assertion. Modern neighbours contribute equally essential constraints: Toulmin on warrants, Schaffer on contrast, Manski on identification under assumptions, Dretske and measurement theory on channel/model conditions, Blackwell and sufficiency on information-preserving processing, Lackey on non-simple transmission, and provenance infrastructures on assertion lineage. The Readout Condition is not a replacement for any of them. It is the architecture produced when their partial lessons are made simultaneously binding at the level of individual claim distinctions.

Its three practical questions remain simple:

$$\boxed{\text{What does the attributed source distinguish?}} \tag{27}$$

$$\boxed{\text{Which distinction does the claim add?}} \tag{28}$$

$$\boxed{\text{Which provenance path makes that addition licit?}} \tag{29}$$

The paper's governing maxim can therefore be retained without qualification:

$$\boxed{\begin{array}{c}\textbf{No epistemic discrimination}\\ \textbf{without provenance.}\end{array}} \tag{30}$$

Author Note

The author is solely responsible for the arguments, formalizations, interpretations, and citations in this manuscript. Generative AI tools were used for literature mapping, adversarial review, LaTeX assistance, and editorial restructuring; the author selected the research direction, checked the formal claims and sources, revised the arguments, and endorses the final manuscript. Public Readout Universe repository artifacts are cited as intellectual and operational lineage, not as independent validation of the theory.

A. Formal Notes

A.1 Why the deterministic condition is local

The factorization proposition requires no global possible-world space. For any local contrast set $\mathcal{X}_Q$ fixed by a question Q, the condition is evaluated only over that domain. If Q changes, the relevant $\mathcal{X}_Q$ may change. This makes the audit contrast-sensitive without making contrast selection arbitrary: exclusions that alter the question's constitutive alternatives must be separately justified.

A.2 Pointwise relational reduction

If deterministic $R : \mathcal{X} \to Y$ induces $xI_Rx' \iff R(x) = R(x')$, then

$$N_R(x_0) = R^{-1}(R(x_0)). \tag{31}$$

The pointwise condition $N_R(x_0) \subseteq \Phi^{-1}(\Phi(x_0))$ for every x_0 is therefore exactly fiber constancy. Conversely, fiber constancy implies the pointwise condition at every centre. No transitive closure is required. For genuinely non-transitive I_R, the condition is evaluated at the actual centre and is not iterated to neighbours.

A.3 Distinction-level attribution map

For a claim token at x_0, the audit domain $D_\Phi(x_0)$ from (3) contains all rival pairs the claim excludes. An attribution map

$$\widehat{\lambda} : D_\Phi(x_0) \to 2^V \tag{32}$$

records the nodes represented as supporting each distinction. Correct attribution requires every credited access node to pass its appropriate local licensing test and every omitted essential dependency to be non-load-bearing. The provenance DAG is thus strictly more informative than a single citation list attached to the whole proposition.

A.4 Blackwell monotonicity and additional access

Let $K_2 = G \circ K_1$. Then K_2 is a garbling of K_1 and cannot dominate K_1 in every decision problem unless the experiments are Blackwell-equivalent. By contrast, a joint experiment $J : (Y_1, Y_2) \mid \mathcal{X}$ can strictly dominate its marginal K_1 because the projection from J to Y_1 is itself a garbling. This formal difference matches the distinction between post-processing and access augmentation.

A.5 Stochastic reduction to the deterministic case

Let a deterministic readout $R : \mathcal{X} \to Y$ induce the kernel $K_R(y \mid x) = \mathbf{1}[y = R(x)]$, and let a deterministic claim $\Phi : \mathcal{X} \to Z$ induce $C_\Phi(z \mid x) = \mathbf{1}[z = \Phi(x)]$. If $C_\Phi = G \circ K_R$, then for every y in $\mathrm{im}(R)$ the kernel $G(\cdot \mid y)$ must concentrate on a single value shared by all x with $R(x) = y$. Hence Φ is constant on the fibers of R and factors as $\Phi = \bar{g} \circ R$. The stochastic condition therefore extends rather than replaces the deterministic source-only licensing condition.

B. Canonical Terminology

Readout
: The distinctions preserved or made available under a specified access relation. The map/relation/channel is the readout structure; a realized y is a readout value. The terms are not interchangeable.

Claim distinction
: For an actual alternative x_0, a rival pair $d_{0x'} = (x_0, x')$ excluded by the actual claim token.

Source-relative licensing
: Entitlement to credit a specified epistemic source with a specified claim distinction.

Pointwise relational licensing
: The condition $N_R(x_0) \subseteq \Phi^{-1}(\Phi(x_0))$ at an actual claim token.

Access admissibility
: In stochastic settings, the condition that a downstream channel requires no target-sensitive route beyond K^*; Blackwell garbling tests this but does not audit hidden assumptions.

Access model
: A claimant's representation $\widetilde{R}$ or $\widetilde{K}$ of the actual access structure R^* or K^*.

Access augmentation
: Addition of a new target-sensitive route.

Contrast/relevance operation
: A change in the alternatives under audit, such as $\mathcal{X} \mapsto \mathcal{X}'$.

Inferential commitment
: A prior, model, theory, bridge, calibration assumption, or other rule applied to an existing basis.

Decision-policy augmentation
: A loss, threshold, utility, or institutional rule mapping epistemic states to actions.

Identification level
: The weakest declared augmentation layer at which a claim distinction becomes excludable/identified.

Typed provenance DAG
: A dependency graph whose nodes are typed by epistemic role and whose root is an individual claim distinction.

Epistemic overreach
: Failure of provenance existence, correct attribution, or essential disclosure.

Silent lift
: A mismatch between actual and represented essential provenance, especially the representation of an augmented distinction as source-generated.

Defeater routing
: Propagation of source-specific defeat along the distinctions that depend on the defeated provenance node.

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